# Title Slide

1. A Secure Document Sharing Platform for Non-Tech Users
2. Quadskill
3. Securing Document Handling for Non-Tech-Savvy Users
4. Cyber hackathon
5. 25/06/25

# Team Information

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Problem Statement Context

# problem

**Elderly and rural people** (non tech-savy people) often need to share or print important documents like **aadhar**, **medical records**,**exam forms**, **property documents** that leads to data breaches, unauthorised access and the misuse or exploitation of personal information

- **lack of encription and secure transmission**
- **No user friendly for non tech savy people**
- **Lack of documents tracking and accountibilty**
- **Document permanently left behind**

Background and significance

In India large portion of population **lack digital literacy**. when they need to share or print document they rely on **whatsapp or usb cable**,
These methods are **insecure, non-traceable, and prone to data leaks.**

As India pushes for Digital India, we have faced a major gap:

- "how we ensure data privacy to our people who cant even know how to use phones"
- The risk is not just inconvenience — it leads to identity theft, medical privacy violation, and financial fraud.
- most of our population is currently digitally dependent on others — that makes them vulnerable.

- 68% of rural Indians rely on others for digital tasks like uploading or printing documents. (MeitY, 2023)
- Only 38% of rural internet users feel confident handling official digital documents.
- 7+ crore citizens still use cyber cafés or public PCs for printing sensitive documents.
- 72% of users have shared private documents via insecure methods like WhatsApp or USB. (DSCI Survey)
- In 2021, over 1,000 Aadhaar numbers were leaked through public systems or file transfers. (UIDAI Reports)

# Proposed Solution: Overview

Our solution lets user to upload and share sensetive documents like (aadhar,medical certificate, exam forms) securily and **easily without needing logging and technical knowledge**

## Workflow:-

- File is encripted in browser(AES-256) for privacy
- users share document via **OTP/QR** or **expiring secure links**
- No login or app required **designed for non tech-savy user**
- acess is logged(ip,time) and **file auto-deleted** after each uses
- Build for low **rural and tech enviroment** (cyber cafes in rural areas)
- **Simple UI**, **big buttons**, **Hindi/English support**

## code concept

### Frontend(NEXT JS + TAILWIND)

Generate **OTP/QR** and **secure links**, encription using **Web cripto API**

### Client side encription

**file encripted** before sending to server,uses **file reader** and **ensure privacy** even if server is compramised

### Backend (NEXT JS + MONGODB)

**/api/upload** ->saves encripted file +metadata(**OTP +expiring**) store logs:**IP, times** and **otp acess**

### Traceability

every file access is **logged**, uploader can see **what, where** and **how** it was used

## Novelty and Innovation

- Browser based encription
- OTP/QR based one time access
- sensetive content detection
- No app or login required
- Access login and auto deletation
- ESpecially built for bharat

Simple UI (for rural and elder person), bilingual text (Hindi + English), no technical difficulties and trustworthy

# Proposed Solution:

our system is a secure **document sharing** system for **non tech savy users** and **elder people**,it **encrypt file** in browser, share it via **OTP/QR** or **secure link** and **auto delete** after one access or 3days , making it a **privacy first** printing and sharing solution

1. user upload document
   - click upload docs
   - choose a file

2. File is encripted in browser
   - Uses web crypto API
   - file is encripted before uploadingl
   - No one can read file (even servers)

3. sensetive content is detected
   - if something unusual dtected show warning
   - scans document and warn if threat
   - labelled red(high security), yellow(moderate risk),green (no risk)

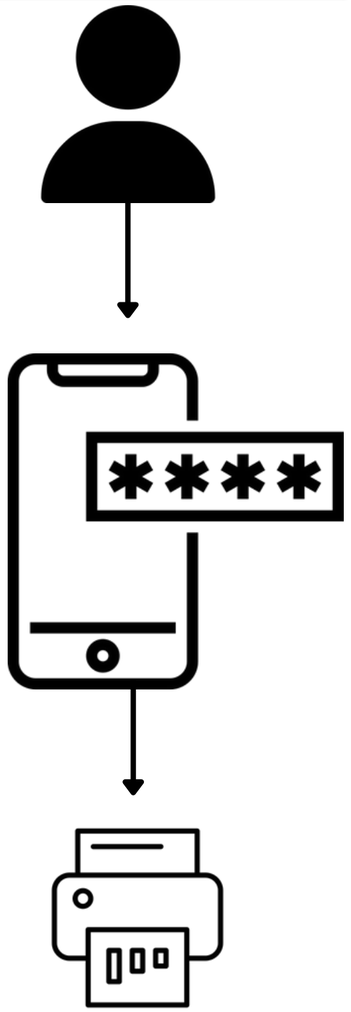
4. secures access generated
   - user choose how to share(otp/qr or link)
   - can set download/print or privacy option for reciever

5. User share
   - no actual file is sent, we have sent only access key

6. Reciever access file
   - open website,enter otp/qr and recieve file
   - prevent download and print unless permitted

7.file auto-deleted and acessed logged
   - every action (view,print,download ) is recorded for tracing of docs
   - IP address, time ,OTP is logged

at the end user has control of docs



# Technology Stack

## Frontend (User Interface)

- **Next.js** – This helped us build the full website easily and in one place. It supports both frontend and backend.
- **Tailwind CSS** – We used it to design clean, mobile-friendly pages with big buttons and simple layout for non-tech users.
- **React Hooks** – For managing file upload, OTP display, etc.

Security

- **WebCrypto API** (AES-256) – This encrypts the document in the browser before uploading. So even our server can not read it.
- **FileReader API** – Reads the file to prepare it for encryption.
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## Backend (Behind the scenes)

- **Next.js API Routes** – We used this to handle file uploads and verify OTPs, and serve files — all built into the same project.
- **Multer** – Helps us securely receive the file from the user and upload it.
- **MongoDB** – Stores OTPs, file expiry info, and access logs.

Sharing Features basicially a db

- **QR Code Generator** – Generates a QR code for easy access in cyber cafe or by doctors.
- **Custom OTP Generato**r – Creates a 6-digit code to access the file safely — no need for login or app.

File Storage

- **Local /tmp or multer** – We temporarily store the encrypted file and delete it after it's used once or after 3 days.

# Innovation and Impact

## innovative aspect

- client side encription before uploading

  files are encripted inside the browser even our server cant see it

- OTP and QR based one time access

  instead of sharing files by unsafe medium like whatsapp,email or usb we only sent a key to receiver

- No App or login needed

  it make it perfect choice for rural and elder person

- No other app can provide this level of security and tracebility

- sensetive docs detection with visual warning

- **whatsapp** disclose the user phone number that can harness privacy of a person and we cant trace the documents
- **DocuSign** is too complex and expensive
- **Dropbox** isn't secure enough for sensitive documents
- our system have unique and more real world features like tracebility, audit,especially designed for low tech people and can easily stand up in the market because lack of solutions

- **Empowers rural** and elderly citizens to safely handle digital documents
- **Restores privacy, dignity, and control** to users who often rely on strangers (like cyber
- **Reduces document misuse** and fraud — saving costs related to identity theft, fake forms, and unauthorized access.
- **Saves time and money** for users who would otherwise travel repeatedly or re-submit lost documents.
- Encourages the use of **affordable digital solutions** over expensive offline paperwork.
- Can c**reate local job opportunities** (e.g., training cyber cafés to use our system responsibly).
- Files are **deleted automatically** after use, avoiding long-term data storage waste.

# Conclusion

## value our system provide

- Puts control of their document to the user this helps all person in India and world , it can protect data breaches, increase trust on online medium
- protect sensetive document for eg. a grandmaa dont be depended on other people she can share their medical report to doctor by their own with full confidence
- works in low tech enviroment and low tech connectivity areas
- it can also create jobs in rural area
- even Indian government can use this system for their work we should not rely on foreign systems

our system is not just a tool — it's a bridge.
A bridge between people and privacy.
Between technology and trust.
Between India's future and those still left behind.